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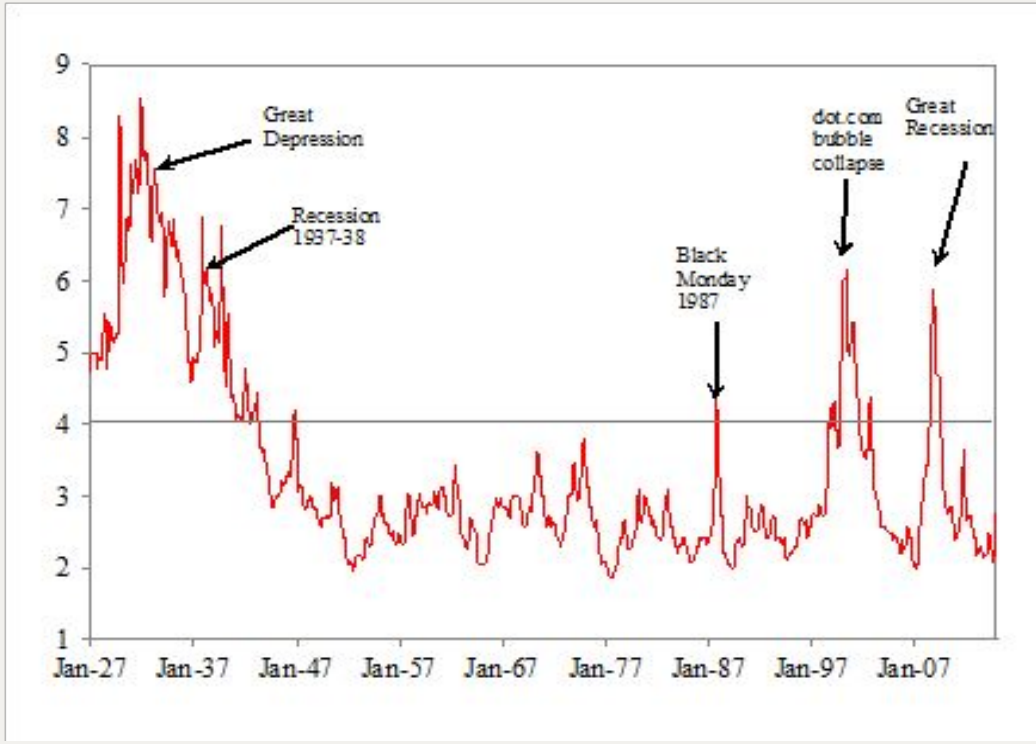
Financial & Risk Analysis

Bayesian Generalized Pareto Modeling of Extreme Stock Market Crashes

Project Context and Motivation



Title: Randomized EVT-Based Crash Prediction

Domain: Finance and Risk Analysis

Objective: To predict the extreme financial losses or tail risk

Approach: Extreme Value Theory (EVT)

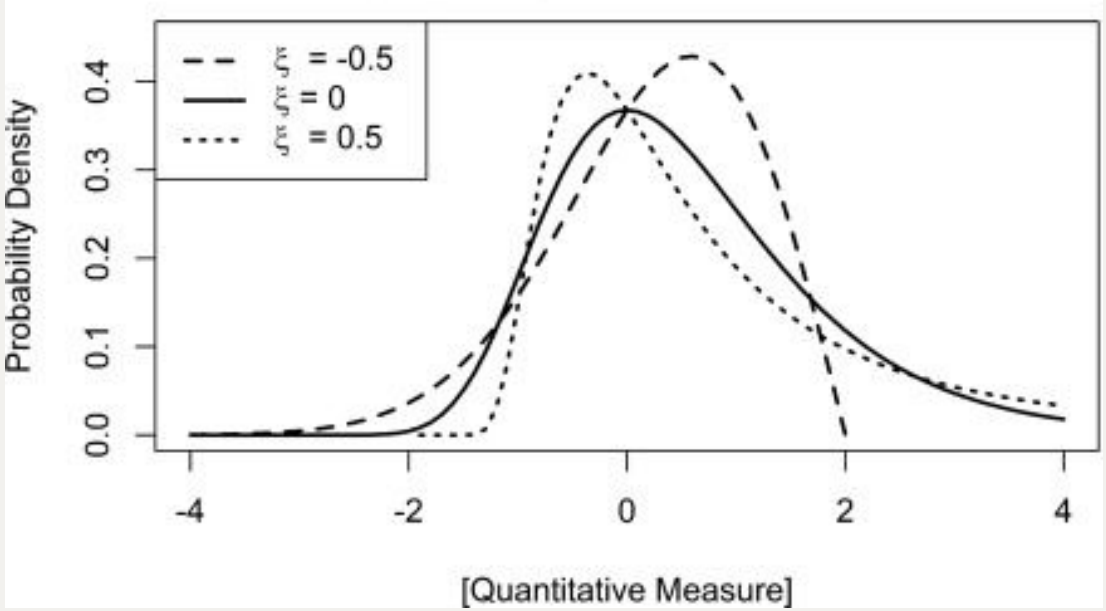
Challenge: High uncertainty in financial markets

Motivation: The need to have probabilistic or even randomized modeling.

Deterministic Baseline Overview

- EVT using **Generalized Pareto Distribution (GPD)**
- The above threshold modeled exceedances.
- Scale parameter:

$$\sigma_i = \exp(x_i^T \beta)$$



- Parameters estimated through the **Maximum Likelihood Estimation (MLE).**
- Deterministic: one setled estimate
- Limitation: overfitting, unstable tail measures

System Pipeline Overview

End-to-End Simulation Pipeline

1. Market Data
2. Feature Engineering (Returns, Volatility)
3. Threshold Exceedance Detection
4. EVT Modeling (GPD Regression)
5. **Parameter Estimation (MLE/MAP)**
6. Crash Risk Prediction

Sources of Uncertainty

- Randomness in the market
- Time-varying volatility
- Sparse extreme observations

Data Flow

Returns

⇩

Feature Extraction (Volatility)

⇩

Extreme Losses (Exceedances)

⇩

GPD Model

⇩

Risk Prediction

Sources of Uncertainty

Data and Measurement Noise

- Errors in computation of returns(log/simple returns)
- Estimating the volatility using uncertainty (EWMA, GK, etc.)

Environmental Variability

- Sudden shocks in the market (crashes, crises)
- Events that are macroeconomic and policy-driven.

Model and Data Uncertainty

- Availability of limited tail data.
- Threshold selection sensitivity (u).
- The model, it's misspecification and parameter instability.

Motivation: Issues with probabilistic / randomized modeling

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Random Variables in the System

Random variables being treated as parameters:

- Regression coefficients β
- Shape parameter ξ

Prior distributions used:

- Gaussian (Ridge)
- Laplace (Lasso)
- Cauchy
- g-prior

Capturing the uncertainty in the estimation.

Randomized Algorithm Design

- Replace MLE with **MAP estimation**.

$$\hat{\beta} = \arg\max \left[\log L(\beta) + \log p(\beta)\right]$$

- Combines likelihood + prior
- Regularization through priors
- Hyperparameters through the cross-validation
- Multiple simulation runs

Probabilistic Reasoning

- Based on **Bayesian inference**

$$p(\beta \mid D) = \frac{p(D \mid \beta)\, p(\beta)}{p(D)}$$

- Balances data and prior knowledge
- Reduces variance in estimates
- Prevents overfitting
- Ensures stability in the tail modeling.

Implementation in Pipeline

- Added MAP estimation functions
- Priors implemented: Ridge, Lasso, Cauchy, g-prior
- Hyperparameters tuned into cross-validation.
- Simulation loop (multiple runs)
- Feature expansion (volatility measures)
- Rest of pipeline unchanged

Results and Performance Analysis

- **RMSE** (Prediction Error):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

- **AIC** (Model Fit):

$$\text{AIC} = 2k - 2\log L$$

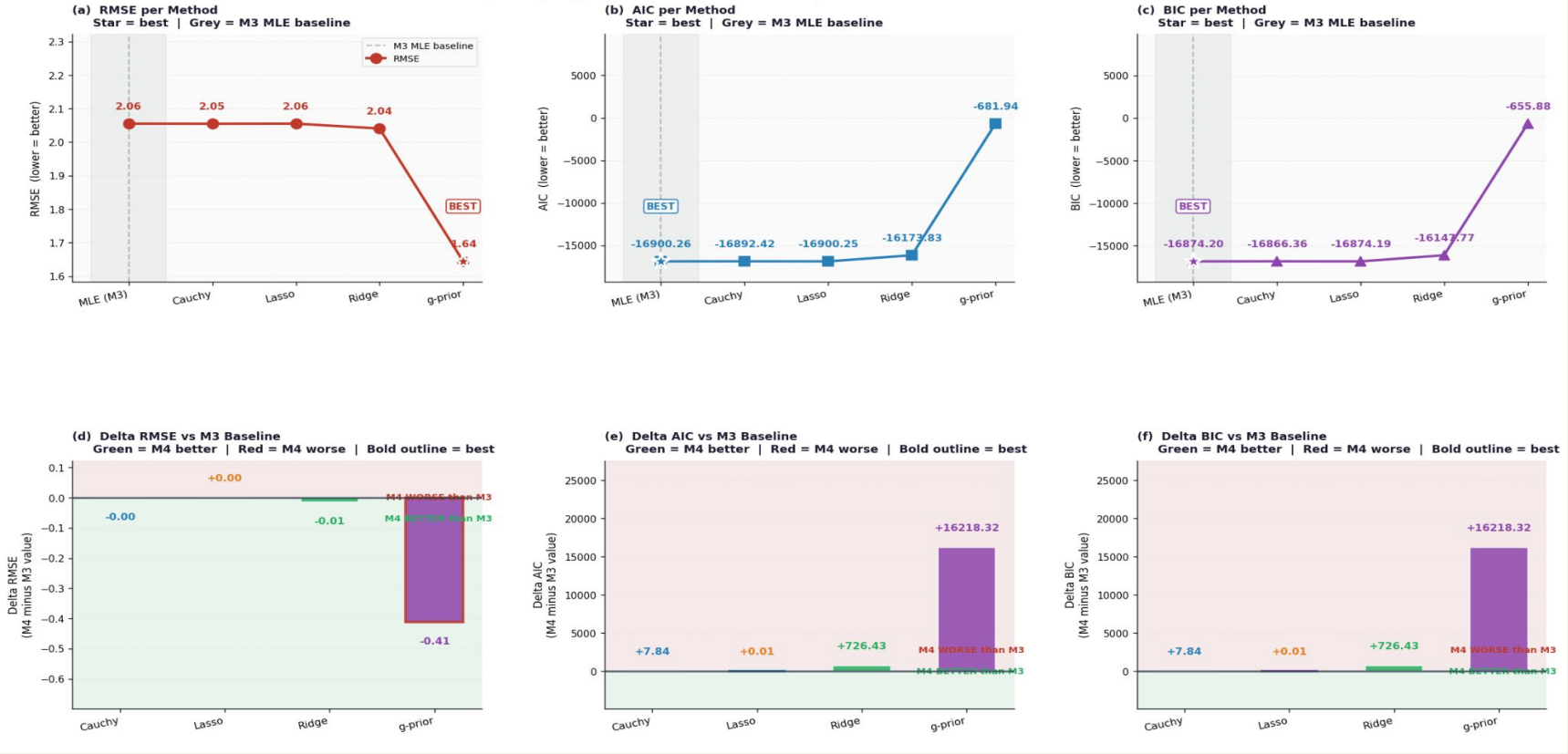
- **BIC** (Model Complexity):

$$\text{BIC} = k\log n - 2\log L$$

- Better performance refers to lower values.

Comparative Results

- Compared:
  - Deterministic MLE (M3)
  - Randomized MAP models (M4)
- Metrics compared: RMSE, AIC, BIC
- To improve RMSE for some priors (e.g: g-prior)
- Accuracy vs. Model complexity
- Performance varies across priors



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Key Insights

- Randomness provides stability of the model.
- Priors set its limit to avoid overfitting in tail data
- Many priors produce different set of behaviors
- Optimal performance is there on data conditions
- Randomized method is far better for limited data

Thank You

Section 2 - Group 16

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